1 **D** Transposing the equation: $k = \frac{H}{A\left|\frac{\Delta T}{l}\right|}$

Hence, the units of *k* are $\frac{J\,s^{-1}}{m^2\,K\,m^{-1}} = \frac{N\,m\,s^{-1}}{m\,K} = \frac{kg\,m\,s^{-2}\,s^{-1}}{K} = \frac{kg\,m\,s^{-3}}{K} = kg\,m\,s^{-3}\,K^{-1}$

[
"Rate of" is considered as "per unit time," so the units of 'rate of energy transfer' are J s^{-1}.]
[Note: answers A and B have the unit 'W,' which is not a base unit; these answers do not have to be considered.